X·EasyGo:An interactive journey bridging reality and digital storytelling through location-based AR technology

Group B3-2 Lei.Zhang(2361007) Hanyan.Niu(2362278) Yiduo.Guo(2360180) Qi.Cao(2360908)

Portfolio Demo

✨Abstract

Traditional campus tours often rely on **one-way explanations**, making visitors **passive listeners** rather than **active explorers**. At Xi'an Jiaotong-Liverpool University (XJTLU), this is especially limiting because the university's **student-centered philosophy** is better understood through **movement, interaction, and experience.** Although existing **academic and commercial solutions** have explored **location-based AR**[1][2], **campus navigation**[3], and **playful interaction**[4], they often fail to combine **practical guidance, meaningful place-based learning, and user engagement** in one coherent experience. In response, this project proposes ***X·EasyGo***, an **AR-supported campus tour system** designed for both **students and visitors.** It aims to provide **clear route-first navigation, role-specific landmark content, and lightweight interactive features** to make campus exploration more **intuitive, engaging, and reflective** of XJTLU's **educational identity.**

🔍Requirement Gathering

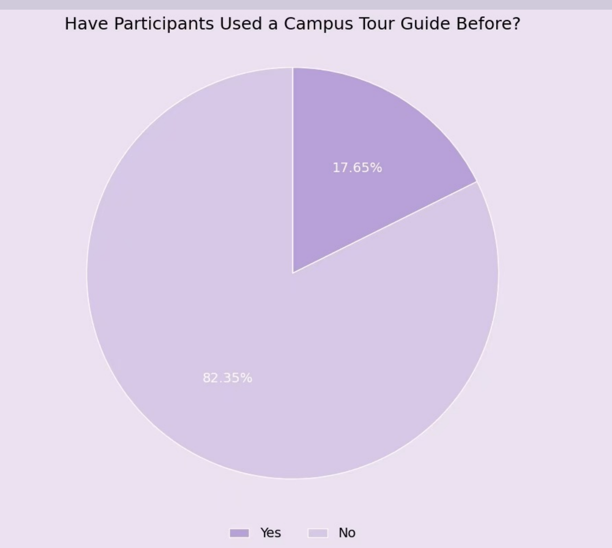

Fig 1. Prior Usage of Campus Tours

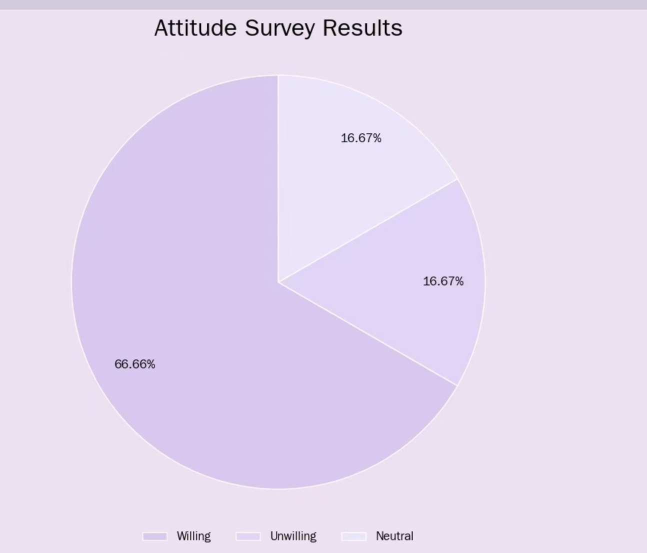

Fig 2. Attitude to the new tour product

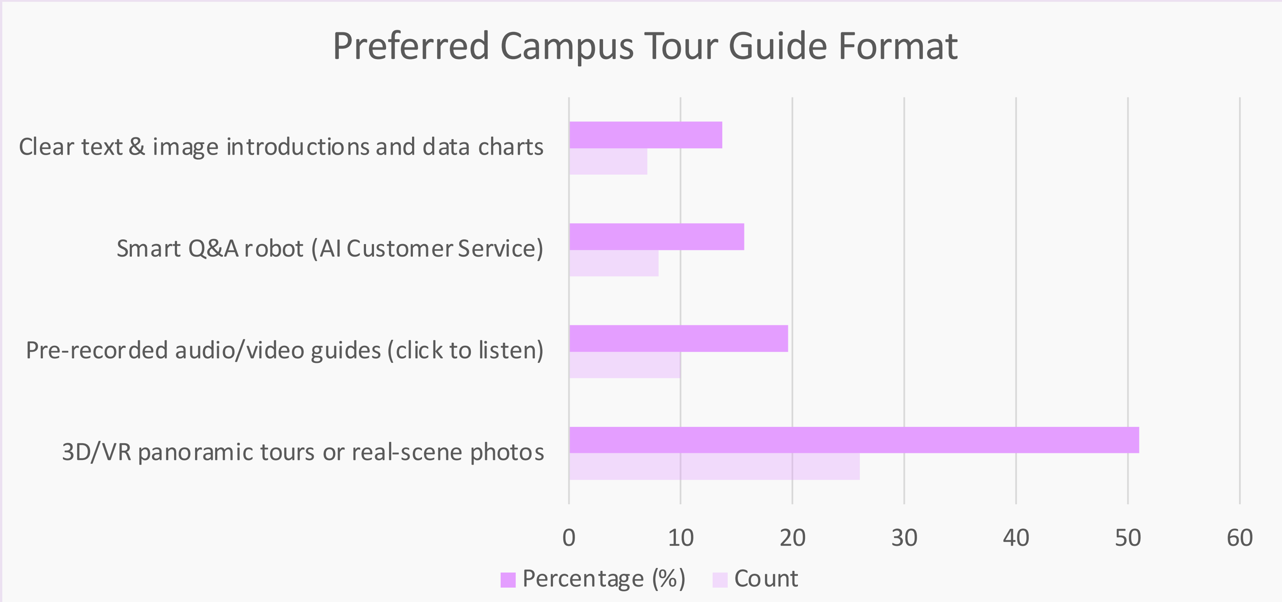

Fig 3. User Preferences for Campus Tour Guide Formats

User Pain Points	Core App Solutions
No customized content for visitors	• Role-specific explanations
Low interaction & engagement	• Gamified travel quests • AR-based guidance
No user-to-user connection	• Message board

Fig 4. Must-to-have functions and details

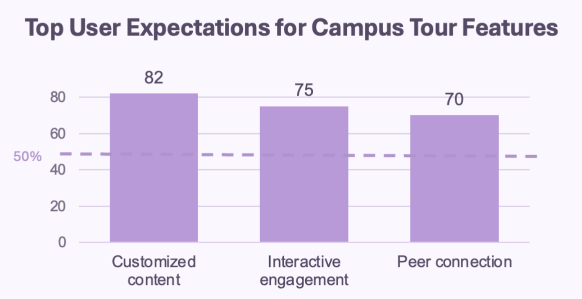

Fig 5. Functionality Recognition

Our investigation (n = 51) reveals that ***82.35%*** of participants have never used existing campus tour services (Fig. 1), while approximately ***66.66%*** expressed a willingness to use such a product to assist them in exploring the campus (Fig. 2). Furthermore, the survey indicates that **most participants prefer VR panoramic tours** as their preferred campus tour format (Fig. 3). In addition, participants expressed a strong expectation for **role-specific content, enhanced interactive engagement, and opportunities** for peer connection (Fig. 5).

Then we make the the ***persona*** shown in Figure 6.

Fig 6. Persona

The ***user journey map*** in Figure 7 illustrates how users interact with our system to address key **pain points.**

Fig 7. User Journey Map

🎨 Prototype Design & Iterations

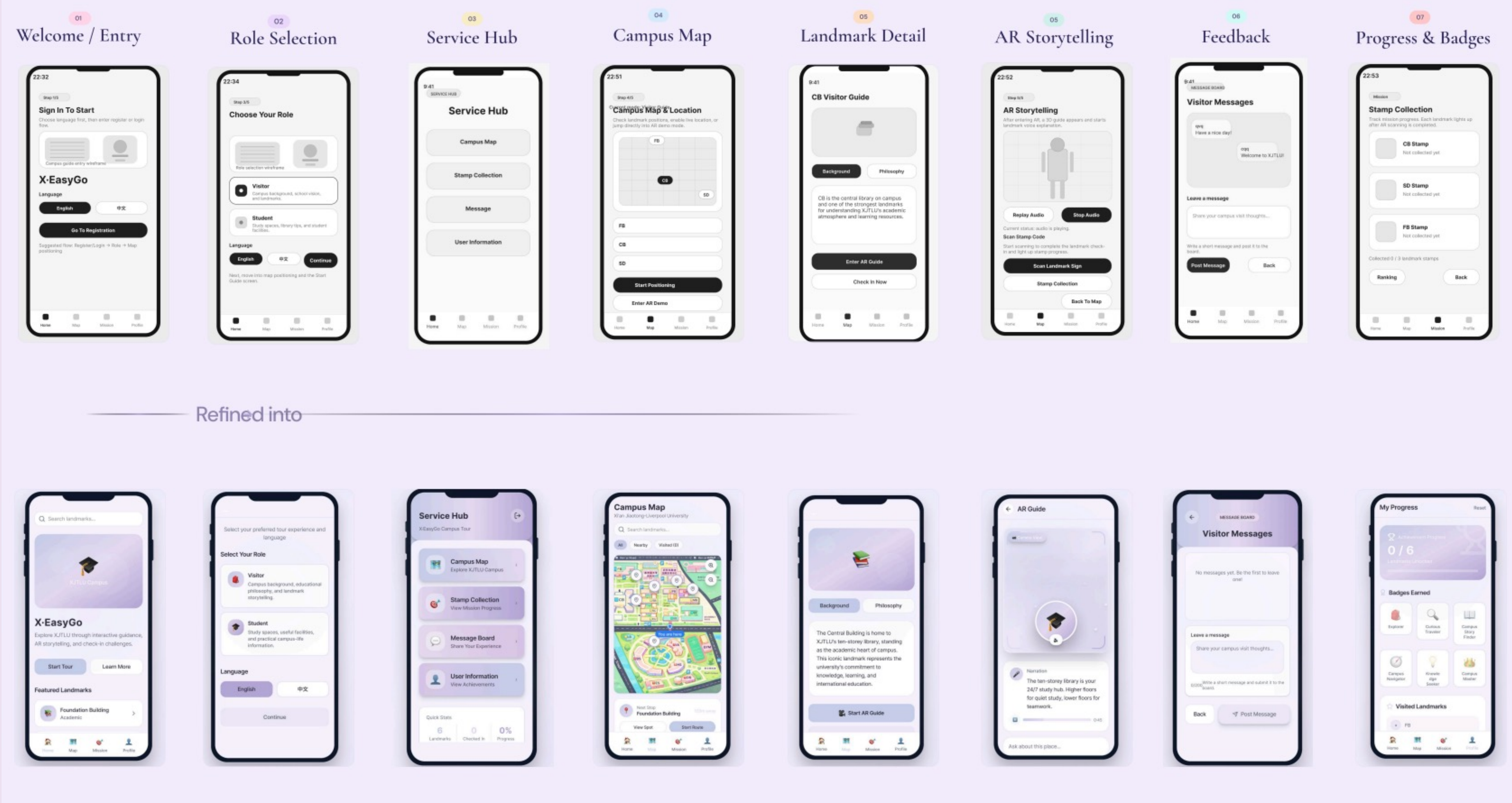

Fig 8. Iterations of app prototype UI design

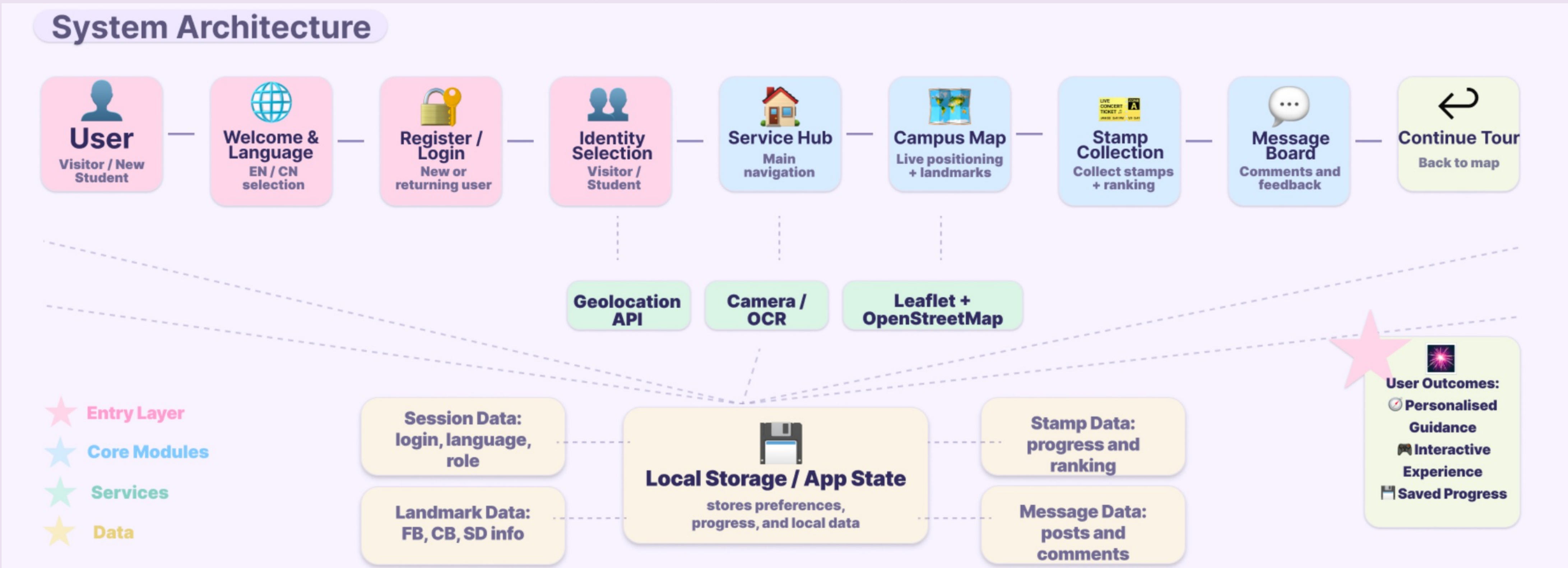

Fig 9. System Architecture

Figure 8 illustrates the iterative design process of the ***app prototype UI***. We first developed a basic version of the system and then progressively added more features and a more **visually appealing interface.** Figure 9 shows the ***system workflow***, illustrating how user interactions move through **core modules, supporting services, and data storage.**

To balance low-friction navigation with playful interaction, we explored **two alternative stamp collection mechanisms**:
(1) an automatic location-based trigger, focusing on efficiency.
(2) a manual AR scanning interaction, emphasizing user engagement.
This design dilemma informed the **development of two prototype variants**, which were later evaluated through user testing.

📊 Evaluation Results

In the evaluation, we invited **10 participants** to use and experience our product.

A/B Testing

Participants (n = 10) were divided into two groups to conduct an A/B test comparing two approaches to stamp collection:

- **Prototype A:** Users automatically receive a stamp when the location system detects that they are near a building.
- **Prototype B:** Users must scan the signboard outside the building to obtain a stamp.

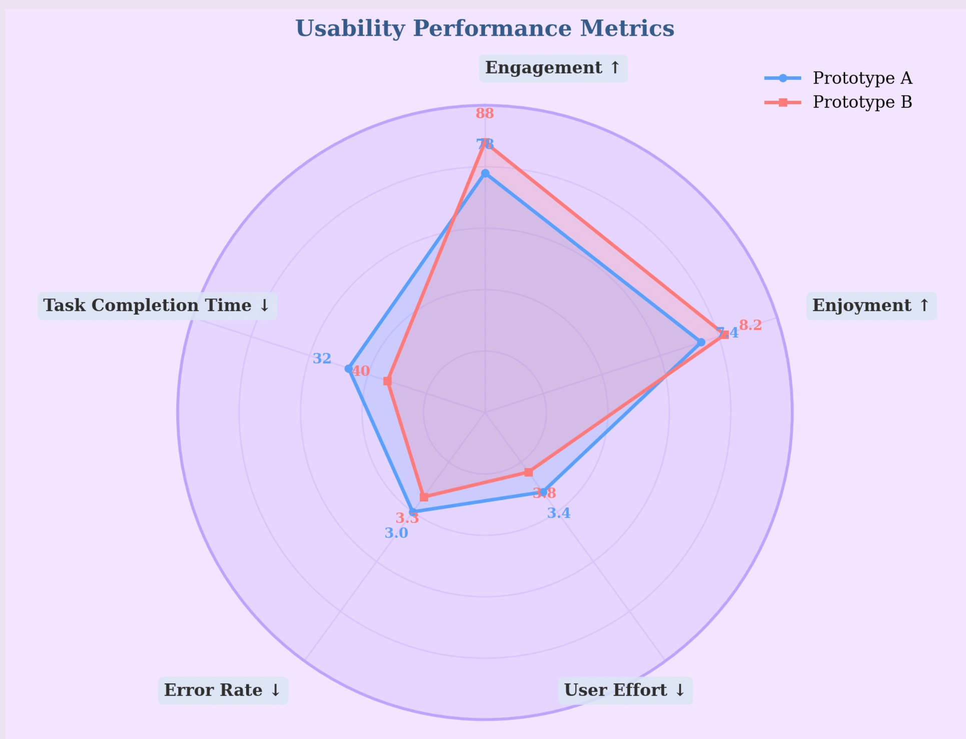

Fig 10. Usability Performance Metric

As shown in Figure 10, **Prototype A** achieves **higher scores** in **task completion time, error rate, and user effort**. However, **Prototype B** performs better in **engagement and enjoyment**, which are key factors in the **user requirements.** Therefore, despite its drawbacks, **Prototype B provides greater overall value to the user experience**. As a result, Prototype B is selected as the final design.

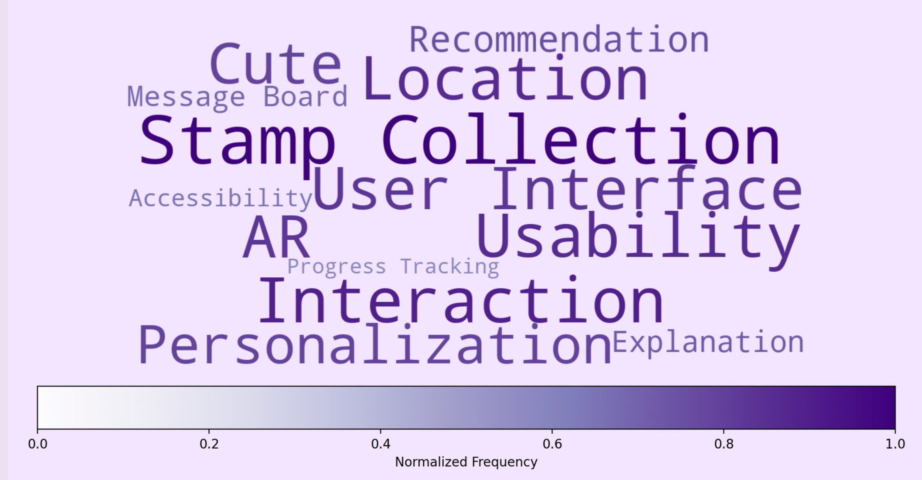

Fig 11. Key Factors Users Value in the Product

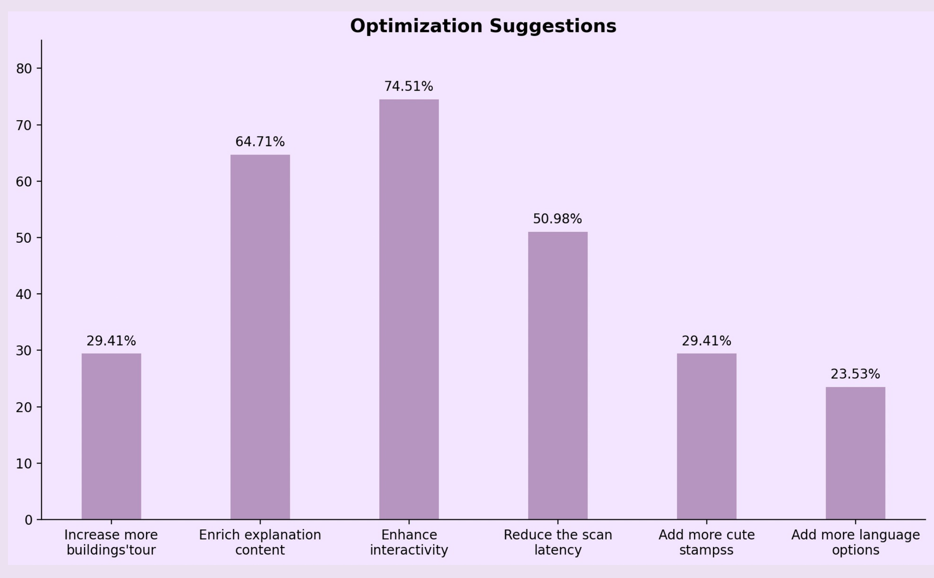

Fig 12. Users' optimization suggestions

Figure 11 presents a **word cloud summarizing key themes** extracted from **users' feedback** after interacting with the system. Notably, **"Stamp Collection"** and **"AR"** appear most frequently, highlighting their importance to users. Figure 12 illustrates **optimization suggestions** provided by participants, where enhancing **interactivity (*74.51%*)** and **enriching explanation content (*64.71%*)** emerge as the most prominent.

🎯 Conclusion

Results suggest that X-EasyGo **improves campus exploration** by combining **clear guidance, role-specific content, and interactive AR features** in a more **engaging and user-centred way.**

Limitations

◆ The AR system is still at **a preliminary stage**, with limited positioning accuracy and delays in the scan-and-collect-stamp function.
◆ The current prototype covers **only three campus buildings**, which limits the scope and completeness of the experience.
◆ The current evaluation is **based on a limited number of participants**, so broader user testing is still needed.

Future Work

◆ Expand the system to **cover more campus buildings and functional areas.**
◆ **Improve the underlying technology** to enhance **positioning accuracy and reduce response delays.**
◆ **Conduct broader testing** with **different user groups** to further refine interaction design and content delivery.

📚 References & Acknowledgments

[1] A. Kleftodimos and A. Evagelou, "Location-based augmented reality in education," *Encyclopedia*, 2025, 5, 54, doi: 10.3390/encyclopedia5020054.

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[4] C.-Y. Hsieh, C.-M. Chen, and Y.-C. Yang, "A game-based augmented reality navigation system to support makerspace user education in a university library," in *2023 14th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI)*, 2023, pp. 218–223, doi: 10.1109/IIAI-AAI59060.2023.00052.